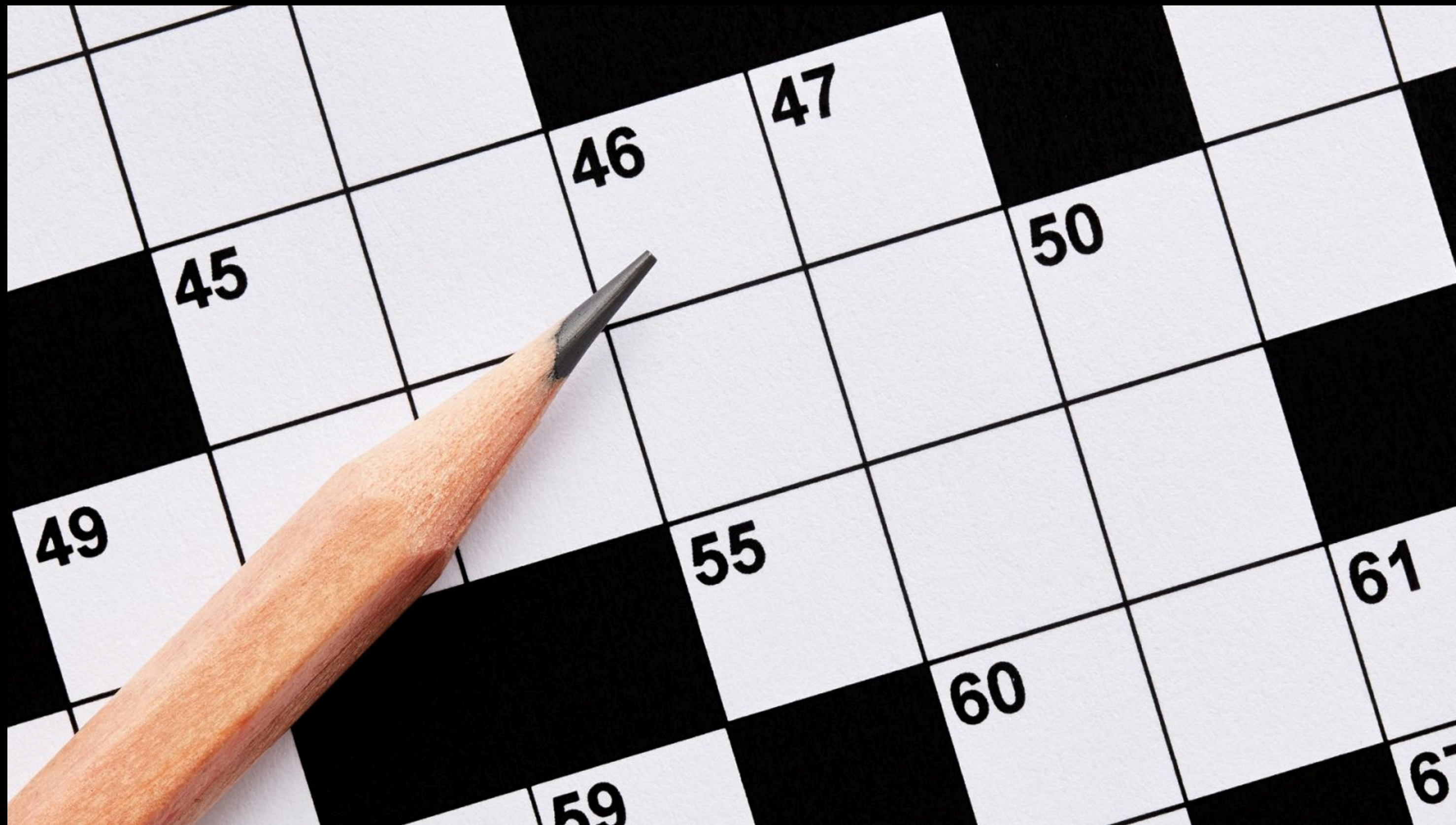




Crucigrama CBD

Tema del proyecto

¿De qué va el proyecto

He hecho una aplicación web para resolver crucigramas.
¿Te cuento cómo lo he hecho?



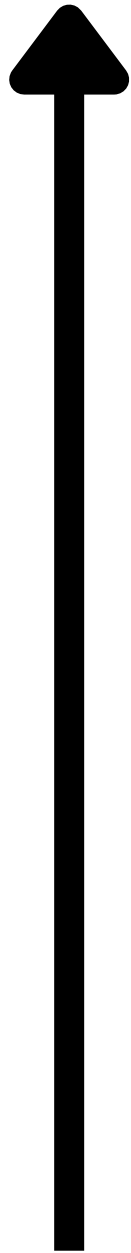
Índice

- Introducción
- Modelo de datos
- Tecnologías
- Conclusiones

Introducción

Cuadrícula

- Filas y columnas
- Celdas negras y blancas



... and this is how it turned out

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					33		34			
		35	36	37		38				
	39						40			

Solutions

ACROSS
1 Rab, 4 Dairy, 8 Uno, 10 Due, 12 Skol, 14 Oat, 16 Pie, 18 Lu, 20 Enol, 22 Ugadi, 25 Tasse, 27 Cops, 29 Ol, 30 Cancellers, 13 Lie, 15 Aug, 17 ENT, 19 Udu, 21 Oak, 23 Action, 24 IPA, 26 Sol, 28 Spa, 31 Tip, 32 Ayes, 34 Rod, 36 NN, 37 EO, 39 All, 33 Airy, 35 One, 38 Poe, 39 Annoy, 40 Ois

DOWN
1 Resolution, 2 Bust, 3 Ink, 5 ID, 6 Au, 7 Rebels, 9 OOP, 11

Across

1 Crop sowing season (4)
4 Quintessential New Year gift (5)
8 Popular card game at house parties (3)
10 Bill waiting to be settled (3)
12 Say 'cheers' like a Viking (4)
14 What's common to a coat and a moat? (3)
16 A dish that is a mathematician's staple (3)
18 A common Goodreads challenge in French (2)
20 I sound like Sherlock's sister but I am a tipsy organic compound (4)
22 Kannada New Year (5)

Down

1 I am made and broken within weeks, sometimes days (10)
2 To stop illegal trading (4)
3 You could use me to write, paint, or modify your body (3)
5 Carry it to show who you are (2)
6 Gold in chemistry (2)
7 Some have a cause, some don't (6)
9 Bengaluru techies need it the most (3)
11 These can jam signals (10)
13 An excuse to avoid an outing (3)
15 By the Gregorian calendar, I will arrive after 212 days in 2023 (3)
17 Expertise of an otolaryngologist (3)
19 I sound like a Hawaiian but I hail from Nigeria. I am a close relative of the Indian percussion instrument ghatam (3)
21 Scrat found his muse under this tree (3)
23 I speak louder than words (6)
24 The British invented me and I am bitter than most ales (3)
26 I am a Mexican beer with a hint of tequila (3)
28 I can beautify and I can heal (3)
31 Compliment for a good meal (3)
32 Sailors' parlance (4)
34 My babies go by the name of a famous chewing gum (3)
36 If you read this paper daily, you know how we abbreviate anonymous (2)
37 A president's power in accordance with the Constitution (2)



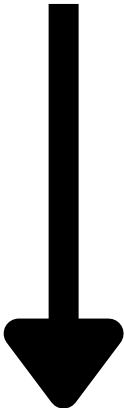
Introducción

Pistas

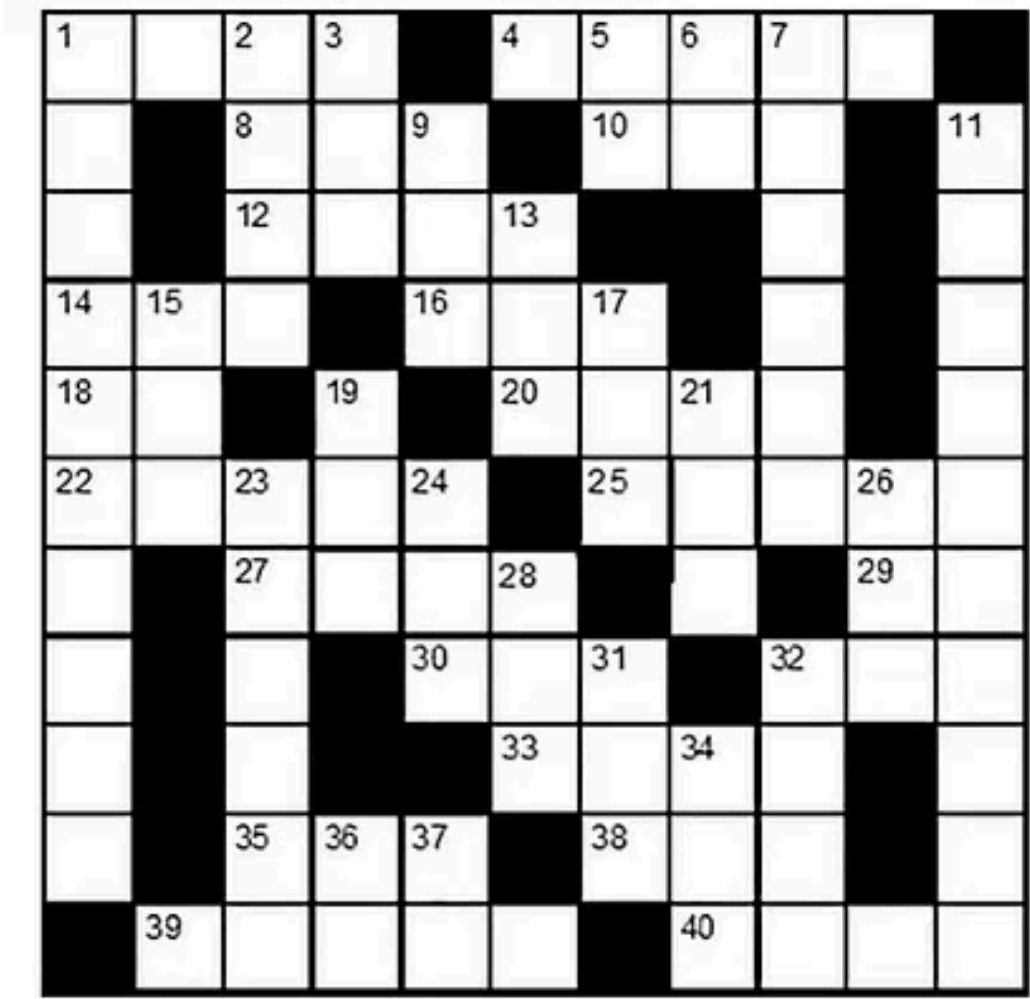
Horizontal



Vertical



... and this is how it turned out



Solutions

ACROSS
1 Rabi, 4 Dairy, 8 Uno, 10 Due, 12 Skol, 14 Oat, 16 Pie, 18 Lu, 20 Enol, 22 Uga, 25 Tasse, 27 Cops, 29 Ol, 30 Cancellers, 13 Lie, 15 Aug, 17 ENT, 19 Udu, 21 Oak, 23 Action, 24 IPA, 26 Sol, 28 Spa, 31 Tip, 32 Ayes, 34 Rod, 36 NN, 37 EO, 39 APL, 32 All, 33 Airy, 35 One, 38 Poe, 39 Annoy, 40 Ois

DOWN
1 Resolution, 2 Bust, 3 Ink, 5 ID, 6 Au, 7 Rebels, 9 OOP, 11

Across

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- 4 Quintessential New Year gift (5)
- 8 Popular card game at house parties (3)
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- 20 I sound like Sherlock's sister but I am a tipsy organic compound (4)
- 22 Kannada New Year (5)
- 25 You can't play beer pong in France without me (5)
- 27 _ and balls: a famous magic trick (4)
- 29 Ancient (2)
- 30 Appropriate or suitable (3)
- 32 A whole lot (3)
- 33 Spacious and ventilated (4)
- 35 Countdowns stop here (3)
- 38 I fall in the same category as Gulzar but Wednesday Addams loves me (3)
- 39 What a child does with pester power (5)
- 40 Suffix to a process or condition (4)

Down

- 1 I am made and broken within weeks, sometimes days (10)
- 2 To stop illegal trading (4)
- 3 You could use me to write, paint, or modify your body (3)
- 5 Carry it to show who you are (2)
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- 37 A president's power in accordance with the Constitution (2)

Modelo de datos

¿Cómo pasamos
un crucigrama
a una aplicación
web?

Hay que pasar
el crucigrama
a un modelo de
datos



Modelo de datos

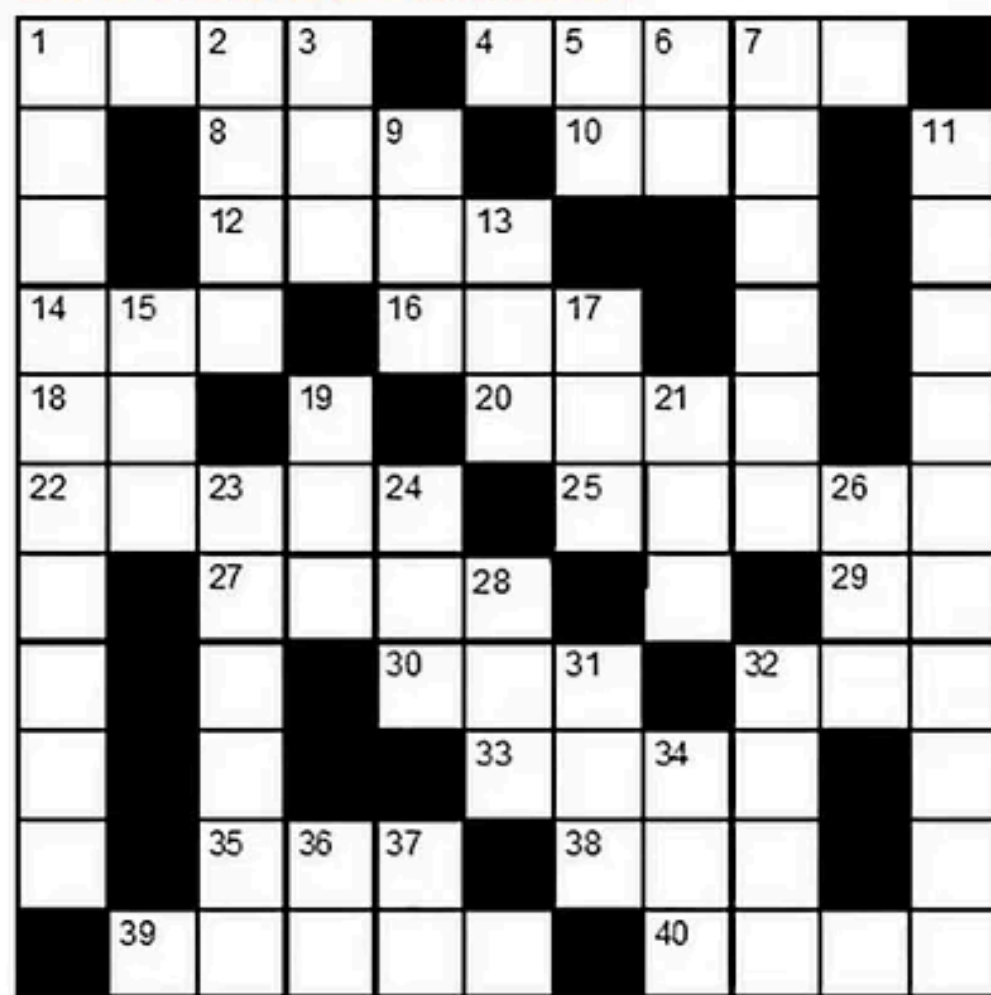
Cuadrícula

Vemos que un crucigrama tiene:

- Cuadrícula
- Celdas
- Pistas

Vamos a modelarlo por partes

... and this is how it turned out

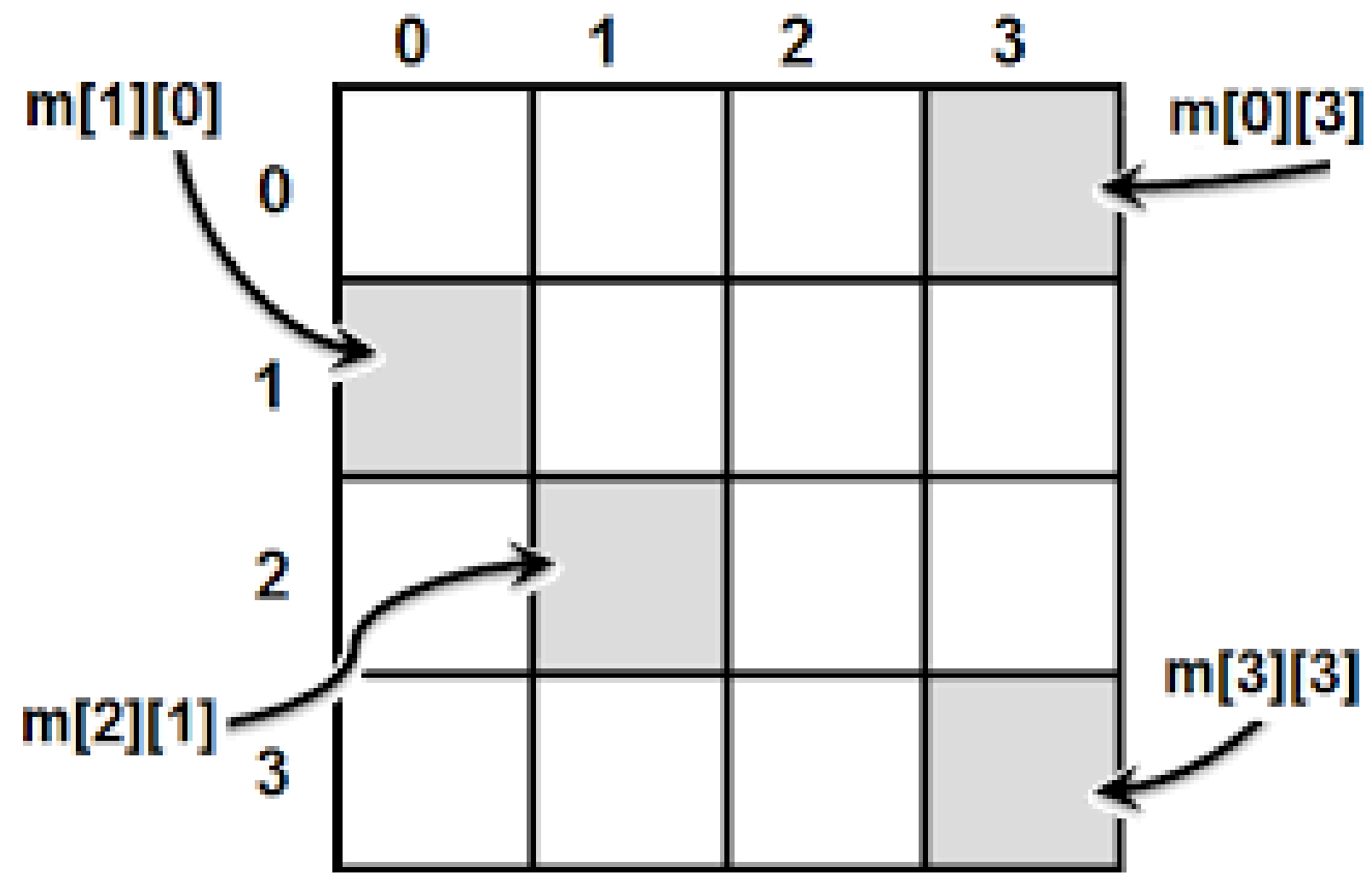


Solutions

ACROSS
1 RABOL, 4 DIARY, 8 UNO, 10 DUE, 12 SKOL, 14 OAT, 16 PIE, 18 LU, 20 ENOL, 22 UGADI, 25 TASSE, 27 CUPS, 29 OI, 30 CANCELLERS, 33 LIE, 35 AUG, 37 ENT, 39 UDU, 41 OAK, 43 ACTION, 44 IPA, 46 SOL, 48 SPA, 51 TIP, 52 AYES, 54 RDO, 56 NN, 57 EO, 58 PA, 60 SOL, 62 SPA, 64 TIP, 66 AYES, 68 RDO, 70 NN, 72 EO, 74 PA, 76 SOL, 78 SPA, 80 TIP, 82 AYES, 84 RDO, 86 NN, 88 EO, 90 PA, 92 SOL, 94 SPA, 96 TIP, 98 AYES, 100 RDO, 102 NN, 104 EO, 106 PA, 108 SOL, 110 SPA, 112 TIP, 114 AYES, 116 RDO, 118 NN, 120 EO, 122 PA, 124 SOL, 126 SPA, 128 TIP, 130 AYES, 132 RDO, 134 NN, 136 EO, 138 PA, 140 SOL, 142 SPA, 144 TIP, 146 AYES, 148 RDO, 150 NN, 152 EO, 154 PA, 156 SOL, 158 SPA, 160 TIP, 162 AYES, 164 RDO, 166 NN, 168 EO, 170 PA, 172 SOL, 174 SPA, 176 TIP, 178 AYES, 180 RDO, 182 NN, 184 EO, 186 PA, 188 SOL, 190 SPA, 192 TIP, 194 AYES, 196 RDO, 198 NN, 200 EO, 202 PA, 204 SOL, 206 SPA, 208 TIP, 210 AYES, 212 RDO, 214 NN, 216 EO, 218 PA, 220 SOL, 222 SPA, 224 TIP, 226 AYES, 228 RDO, 230 NN, 232 EO, 234 PA, 236 SOL, 238 SPA, 240 TIP, 242 AYES, 244 RDO, 246 NN, 248 EO, 250 PA, 252 SOL, 254 SPA, 256 TIP, 258 AYES, 260 RDO, 262 NN, 264 EO, 266 PA, 268 SOL, 270 SPA, 272 TIP, 274 AYES, 276 RDO, 278 NN, 280 EO, 282 PA, 284 SOL, 286 SPA, 288 TIP, 290 AYES, 292 RDO, 294 NN, 296 EO, 298 PA, 300 SOL, 302 SPA, 304 TIP, 306 AYES, 308 RDO, 310 NN, 312 EO, 314 PA, 316 SOL, 318 SPA, 320 TIP, 322 AYES, 324 RDO, 326 NN, 328 EO, 330 PA, 332 SOL, 334 SPA, 336 TIP, 338 AYES, 340 RDO, 342 NN, 344 EO, 346 PA, 348 SOL, 350 SPA, 352 TIP, 354 AYES, 356 RDO, 358 NN, 360 EO, 362 PA, 364 SOL, 366 SPA, 368 TIP, 370 AYES, 372 RDO, 374 NN, 376 EO, 378 PA, 380 SOL, 382 SPA, 384 TIP, 386 AYES, 388 RDO, 390 NN, 392 EO, 394 PA, 396 SOL, 398 SPA, 400 TIP, 402 AYES, 404 RDO, 406 NN, 408 EO, 410 PA, 412 SOL, 414 SPA, 416 TIP, 418 AYES, 420 RDO, 422 NN, 424 EO, 426 PA, 428 SOL, 430 SPA, 432 TIP, 434 AYES, 436 RDO, 438 NN, 440 EO, 442 PA, 444 SOL, 446 SPA, 448 TIP, 450 AYES, 452 RDO, 454 NN, 456 EO, 458 PA, 460 SOL, 462 SPA, 464 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RDO, 694 NN, 696 EO, 698 PA, 700 SOL, 702 SPA, 704 TIP, 706 AYES, 708 RDO, 710 NN, 712 EO, 714 PA, 716 SOL, 718 SPA, 720 TIP, 722 AYES, 724 RDO, 726 NN, 728 EO, 730 PA, 732 SOL, 734 SPA, 736 TIP, 738 AYES, 740 RDO, 742 NN, 744 EO, 746 PA, 748 SOL, 750 SPA, 752 TIP, 754 AYES, 756 RDO, 758 NN, 760 EO, 762 PA, 764 SOL, 766 SPA, 768 TIP, 770 AYES, 772 RDO, 774 NN, 776 EO, 778 PA, 780 SOL, 782 SPA, 784 TIP, 786 AYES, 788 RDO, 790 NN, 792 EO, 794 PA, 796 SOL, 798 SPA, 800 TIP, 802 AYES, 804 RDO, 806 NN, 808 EO, 810 PA, 812 SOL, 814 SPA, 816 TIP, 818 AYES, 820 RDO, 822 NN, 824 EO, 826 PA, 828 SOL, 830 SPA, 832 TIP, 834 AYES, 836 RDO, 838 NN, 840 EO, 842 PA, 844 SOL, 846 SPA, 848 TIP, 850 AYES, 852 RDO, 854 NN, 856 EO, 858 PA, 860 SOL, 862 SPA, 864 TIP, 866 AYES, 868 RDO, 870 NN, 872 EO, 874 PA, 876 SOL, 878 SPA, 880 TIP, 882 AYES, 884 RDO, 886 NN, 888 EO, 890 PA, 892 SOL, 894 SPA, 896 TIP, 898 AYES, 900 RDO, 902 NN, 904 EO, 906 PA, 908 SOL, 910 SPA, 912 TIP, 914 AYES, 916 RDO, 918 NN, 920 EO, 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2160 TIP, 2162 AYES, 2164 RDO, 2166 NN, 2168 EO, 2170 PA, 2172 SOL, 2174 SPA, 2176 TIP, 2178 AYES, 2180 RDO, 2182 NN, 2184 EO, 2186 PA, 2188 SOL, 2190 SPA, 2192 TIP, 2194 AYES, 2196 RDO, 2198 NN, 2200 EO, 2202 PA, 2204 SOL, 2206 SPA, 2208 TIP, 2210 AYES, 2212 RDO, 2214 NN, 2216 EO, 2218 PA, 2220 SOL, 2222 SPA, 2224 TIP, 2226 AYES, 2228 RDO, 2230 NN, 2232 EO, 2234 PA, 2236 SOL, 2238 SPA, 2240 TIP, 2242 AYES, 2244 RDO, 2246 NN, 2248 EO, 2250 PA, 2252 SOL, 2254 SPA, 2256 TIP, 2258 AYES, 2260 RDO, 2262 NN, 2264 EO, 2266 PA, 2268 SOL, 2270 SPA, 2272 TIP, 2274 AYES, 2276 RDO, 2278 NN, 2280 EO, 2282 PA, 2284 SOL, 2286 SPA, 2288 TIP, 2290 AYES, 2292 RDO, 2294 NN, 2296 EO, 2298 PA, 2300 SOL, 2302 SPA, 2304 TIP, 2306 AYES, 2308 RDO, 2310 NN, 2312 EO, 2314 PA, 2316 SOL, 2318 SPA, 2320 TIP, 2322 AYES, 2324 RDO, 2326 NN, 2328 EO, 2330 PA, 2332 SOL, 2334 SPA, 2336 TIP, 2338 AYES, 2340 RDO, 2342 NN, 2344 EO, 2346 PA, 2348 SOL, 2350 SPA, 2352 TIP, 2354 AYES, 2356 RDO, 2358 NN, 2360 EO, 2362 PA, 2364 SOL, 2366 SPA, 2368 TIP, 2370 AYES, 2372 RDO, 2374 NN, 2376 EO, 2378 PA, 2380 SOL, 2382 SPA, 2384 TIP, 2386 AYES, 2388 RDO, 2390 NN, 2392 EO, 2394 PA, 2396 SOL, 2398 SPA, 2400 TIP, 2402 AYES, 2404 RDO, 2406 NN, 2408 EO, 2410 PA, 2412 SOL, 2414 SPA, 2416 TIP, 2418 AYES, 2420 RDO, 2422 NN, 2424 EO, 2426 PA, 2428 SOL, 2430 SPA, 2432 TIP, 2434 AYES, 2436 RDO, 2438 NN, 2440 EO, 2442 PA, 2444 SOL, 2446 SPA, 2448 TIP, 2450 AYES, 2452 RDO, 2454 NN, 2456 EO, 2458 PA, 2460 SOL, 2462 SPA, 2464 TIP, 2466 AYES, 2468 RDO, 2470 NN, 2472 EO, 2474 PA, 2476 SOL, 2478 SPA, 2480 TIP, 2482 AYES, 2484 RDO, 2486 NN, 2488 EO, 2490 PA, 2492 SOL, 2494 SPA, 2496 TIP, 2498 AYES, 2500 RDO, 2502 NN, 2504 EO, 2506 PA, 2508 SOL, 2510 SPA, 2512 TIP, 2514 AYES, 2516 RDO, 2518 NN, 2520 EO, 2522 PA, 2524 SOL, 2526 SPA, 2528 TIP, 2530 AYES, 2532 RDO, 2534 NN, 2536 EO, 2538 PA, 2540 SOL, 2542 SPA, 2544 TIP, 2546 AYES, 2548 RDO, 2550 NN, 2552 EO, 2554 PA, 2556 SOL, 2558 SPA, 2560 TIP, 2562 AYES, 2564 RDO, 2566 NN, 2568 EO, 2570 PA, 2572 SOL, 2574 SPA, 2576 TIP, 2578 AYES, 2580 RDO, 2582 NN, 2584 EO, 2586 PA, 2588 SOL, 2590 SPA, 2592 TIP, 2594 AYES, 2596 RDO, 2598 NN, 2600 EO, 2602 PA, 2604 SOL, 2606 SPA, 2608 TIP, 2610 AYES, 2612 RDO, 2614 NN, 2616 EO, 2618 PA, 2620 SOL, 2622 SPA, 2624 TIP, 2626 AYES, 2628 RDO, 2630 NN, 2632 EO, 2634 PA, 2636 SOL, 2638 SPA, 2640 TIP, 2642 AYES, 2644 RDO, 2646 NN, 2648 EO, 2650 PA, 2652 SOL, 2654 SPA, 2656 TIP, 2658 AYES, 2660 RDO, 2662 NN, 2664 EO, 2666 PA, 2668 SOL, 2670 SPA, 2672 TIP, 2674 AYES, 2676 RDO, 2678 NN, 2680 EO, 2682 PA, 2684 SOL, 2686 SPA, 2688 TIP, 2690 AYES, 2692 RDO, 2694 NN, 2696 EO, 2698 PA, 2700 SOL, 2702 SPA, 2704 TIP, 2706 AYES, 2708 RDO, 2710 NN, 2712 EO, 2714 PA, 2716 SOL, 2718 SPA, 2720 TIP, 2722 AYES, 2724 RDO, 2726 NN, 2728 EO, 2730 PA, 2732 SOL, 2734 SPA, 2736 TIP, 2738 AYES, 2740 RDO, 2742 NN, 2744 EO, 2746 PA, 2748 SOL, 2750 SPA, 2752 TIP, 2754 AYES, 2756 RDO, 2758 NN, 2760 EO, 2762 PA, 2764 SOL, 2766 SPA, 2768 TIP, 2770 AYES, 2772 RDO, 2774 NN, 2776 EO, 2778 PA, 2780 SOL, 2782 SPA, 2784 TIP, 2786 AYES, 2788 RDO, 2790 NN, 2792 EO, 2794 PA, 2796 SOL, 2798 SPA, 2800 TIP, 2802 AYES, 2804 RDO, 2806 NN, 2808 EO, 2810 PA, 2812 SOL, 2814 SPA, 2816 TIP, 2818 AYES, 2820 RDO, 2822 NN, 2824 EO, 2826 PA, 2828 SOL, 2830 SPA, 2832 TIP, 2834 AYES, 2836 RDO, 2838 NN, 2840 EO, 2842 PA, 2844 SOL, 2846 SPA, 2848 TIP, 2850 AYES, 2852 RDO, 2854 NN, 2856 EO, 2858 PA, 2860 SOL, 2862 SPA, 2864 TIP, 2866 AYES, 2868 RDO, 2870 NN, 2872 EO, 2874 PA, 2876 SOL, 2878 SPA, 2880 TIP, 2882 AYES, 2884 RDO, 2886 NN, 2888 EO, 2890 PA, 2892 SOL, 2894 SPA, 2896 TIP, 2898 AYES, 2900 RDO, 2902 NN, 2904 EO, 2906 PA, 2908 SOL, 2910 SPA, 2912 TIP, 2914 AYES, 2916 RDO, 2918 NN, 2920 EO, 2922 PA, 2924 SOL, 2926 SPA, 2928 TIP, 2930 AYES, 2932 RDO, 2934 NN, 2936 EO, 2938 PA, 2940 SOL, 2942 SPA, 2944 TIP, 2946 AYES, 2948 RDO, 2950 NN, 2952 EO, 2954 PA, 2956 SOL, 2958 SPA, 2960 TIP, 2962 AYES, 2964 RDO, 2966 NN, 2968 EO, 2970 PA, 2972 SOL, 2974 SPA, 2976 TIP, 2978 AYES, 2980 RDO, 2982 NN, 2984 EO, 2986 PA, 2988 SOL, 2990 SPA, 2992 TIP, 2994 AYES, 2996 RDO, 2998 NN, 3000 EO, 3002 PA, 3004 SOL, 3006 SPA, 3008 TIP, 3010 AYES, 3012 RDO, 3014 NN, 3016 EO, 3018 PA, 3020 SOL, 3022 SPA, 3024 TIP, 3026 AYES, 3028 RDO, 3030 NN, 3032 EO, 3034 PA, 3036 SOL, 3038 SPA, 3040 TIP, 3042 AYES, 3044 RDO, 3046 NN, 3048 EO, 3050 PA, 3052 SOL, 3054 SPA, 3056 TIP, 3058 AYES, 3060 RDO, 3062 NN, 3064 EO, 3066 PA, 3068 SOL, 3070 SPA, 3072 TIP, 3074 AYES, 3076 RDO, 3078 NN, 3080 EO, 3082 PA, 3084 SOL, 3086 SPA, 3088 TIP, 3090 AYES, 3092 RDO, 3094 NN, 3096 EO, 3098 PA, 3100 SOL, 3102 SPA, 3104 TIP, 3106 AYES, 3108 RDO, 3110 NN, 3112 EO, 3114 PA, 3116 SOL, 3118 SPA, 3120 TIP, 3122 AYES, 3124 RDO, 3126 NN, 3128 EO, 3130 PA, 3132 SOL, 3134 SPA, 3136 TIP, 3138 AYES, 3140 RDO, 3142 NN, 3144 EO, 3146 PA, 3148 SOL, 3150 SPA, 3152 TIP, 3154 AYES, 3156 RDO, 3158 NN, 3160 EO, 3162 PA, 3164 SOL, 3166 SPA, 3168 TIP, 3170 AYES, 3172 RDO, 3174 NN, 3176 EO, 3178 PA, 3180 SOL, 3182 SPA, 3184 TIP, 3186 AYES, 3188 RDO, 3190 NN, 3192 EO, 3194 PA, 3196 SOL, 3198 SPA, 3200 TIP, 3202 AYES, 3204 RDO, 3206 NN, 3208 EO, 3210 PA, 3212 SOL, 3214 SPA, 3216 TIP, 3218 AYES, 3220 RDO, 3222 NN, 3224 EO, 3226 PA, 3228 SOL, 3230 SPA, 3232 TIP, 3234 AYES, 3236 RDO, 3238 NN, 3240 EO, 3242 PA, 3244 SOL, 3246 SPA, 3248 TIP, 3250 AYES, 3252 RDO, 3254 NN, 3256 EO, 3258 PA, 3260 SOL, 3262 SPA, 3264 TIP, 3266 AYES, 3268 RDO, 3270 NN, 3272 EO, 3274 PA, 3276 SOL, 3278 SPA, 3280 TIP, 3282 AYES, 3284 RDO, 3286 NN, 3288 EO, 3290 PA, 3292 SOL, 3294 SPA, 3296 TIP, 3298 AYES, 3300 RDO, 3302 NN, 3304 EO, 3306 PA, 3308 SOL, 3310 SPA, 3312 TIP, 3314 AYES, 3316 RDO, 3318 NN, 3320 EO, 3322 PA, 3324 SOL, 3326 SPA, 3328 TIP, 3330 AYES, 3332 RDO, 3334 NN, 3336 EO, 3338 PA, 3340 SOL, 3342 SPA, 3344 TIP, 3346 AYES, 3348 RDO, 3350 NN, 3352 EO, 3354 PA, 3356 SOL, 3358 SPA, 3360 TIP, 3362 AYES, 3364 RDO, 3366 NN, 3368 EO, 3370 PA, 3372 SOL, 3374 SPA, 3376 TIP, 3378 AYES, 3380 RDO, 3382 NN, 3384 EO, 3386 PA, 3388 SOL, 3390 SPA, 3392 TIP, 3394 AYES, 3396 RDO, 3398 NN, 3400 EO, 3402 PA, 3404 SOL, 3406 SPA, 3408 TIP, 3410 AYES, 3412 RDO, 3414 NN, 3416 EO, 3418 PA, 3420 SOL, 3422 SPA, 3424 TIP, 3426 AYES, 3428 RDO, 3430 NN, 3432 EO, 3434 PA, 3436 SOL, 3438 SPA, 3440 TIP, 3442 AYES, 3

Modelo de datos

La cuadrícula es una lista de 2 dimensiones.
Tiene celdas con dos estados



Modelo de datos



¿No faltan las
pistas en la
cuadrícula?



Tienes razón. Además
hay dos
formas de dar pistas

Modelo de datos

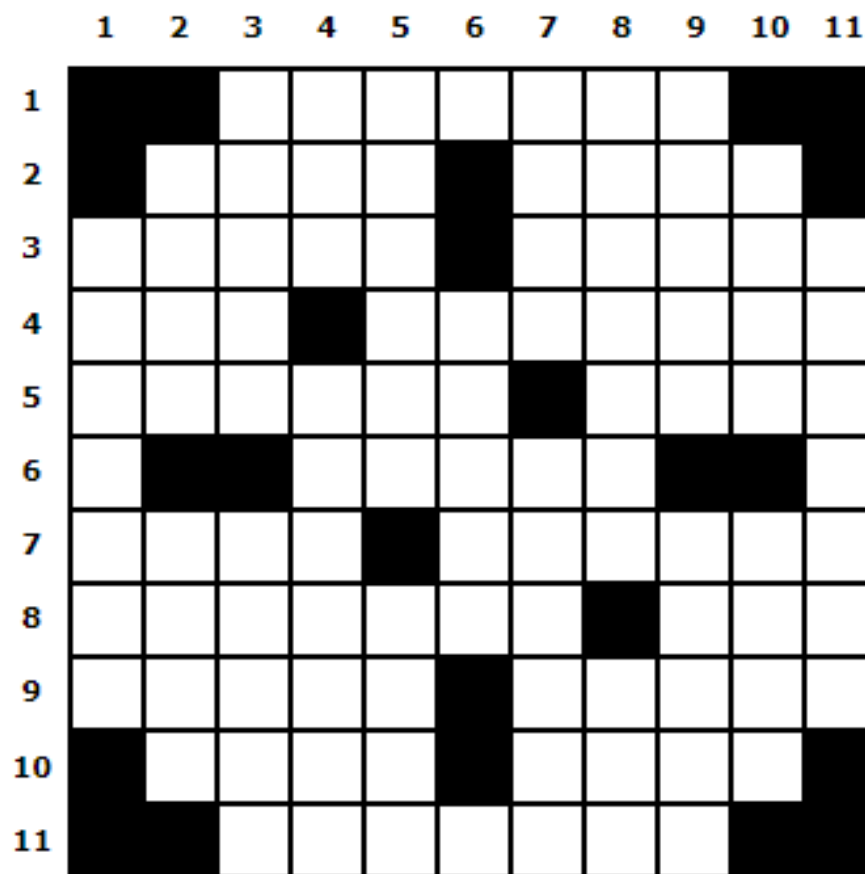
Pistas



Coordenadas



Número en las celdas



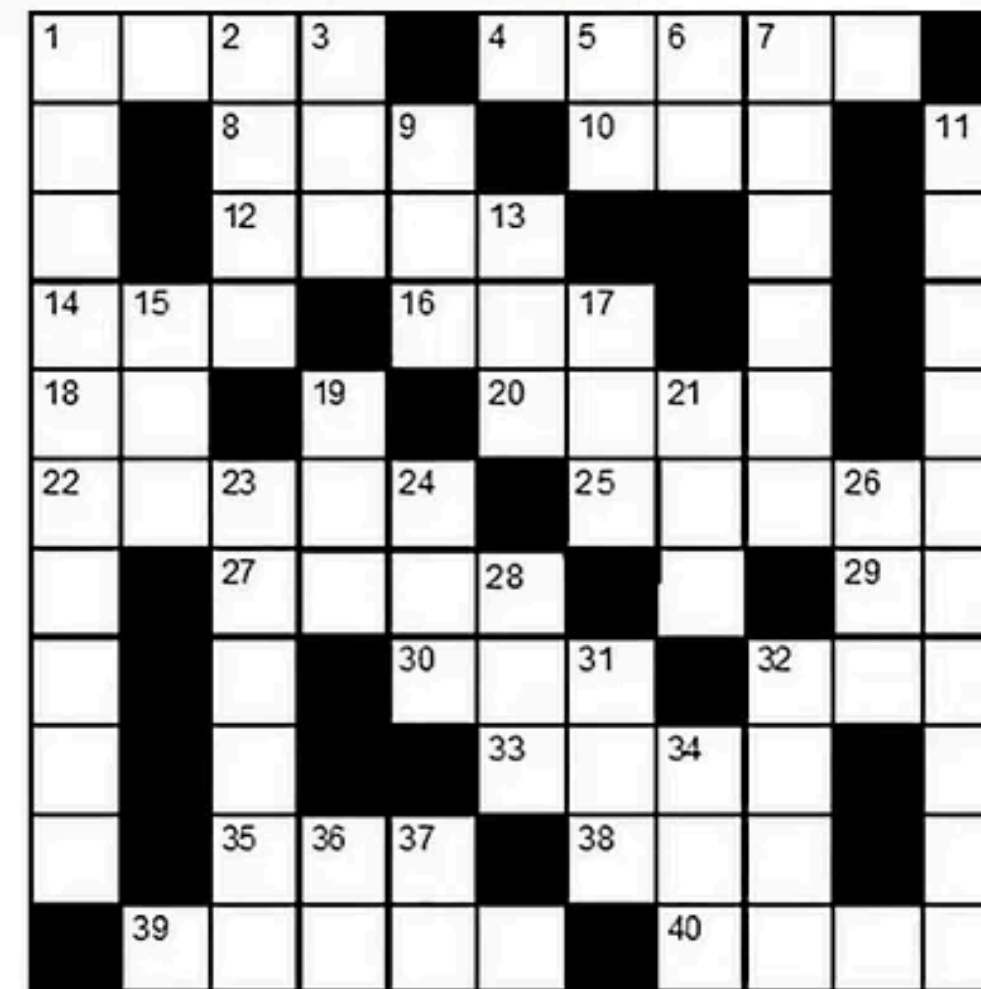
● HORIZONTALES

1: Que tiene dos focos. **2:** Papeleta en la que se da a cor elección. Adorar. **3:** Hombre zafio y ordinario. Ser u objeto animal, que en la mitología de algunas sociedades se tor atreve. Íbera. **5:** Moderada, escasa, limitada. Cociné u Símbolo del uranio. Bobina, carrete. Cincuenta en romanos de algo. Del disco intervertebral. **8:** Perteneciente o relati Seísmo. Prenda informal, con respecto a la de vestir. **10:** l Asa, parte por donde se coge alguna cosa.

○ VERTICALES

1: Males. **2:** Recipiente para beber. Capital de Samoa. **3:** que en química y en mineralogía forma nombres de mir lugar. **5:** Perteneciente o relativo a la voz o al sonido. Puse Símbolo del oxígeno. Cubo, barreño. Consonante. **7:** Suspe Que tienen forma irregular. Preposición que denota multip Prefijo que significa 'tiempo'. **10:** Ore. Prefijo que significa :

... and this is how it turned out



Solutions

ACROSS
1 Rabol, 4 Diary, 8 Uno, 10 Due, 12 Skol, 14 Oat, 16 Pie, 18 Lu, 20 Enol, 22 Ugadi, 25 Tasse, 27 Cups, 29 Ol, 30 Cancellers, 13 Lie, 15 Aug, 17 ENT, 19 Udu, 21 Oak, 23 Action, 24 IPA, 26 Sol, 28 Spa, 31 Tip, 32 Ayes, 34 Rdo, 36 NN, 37 EO, 39 Annoy, 40 Osis

Across

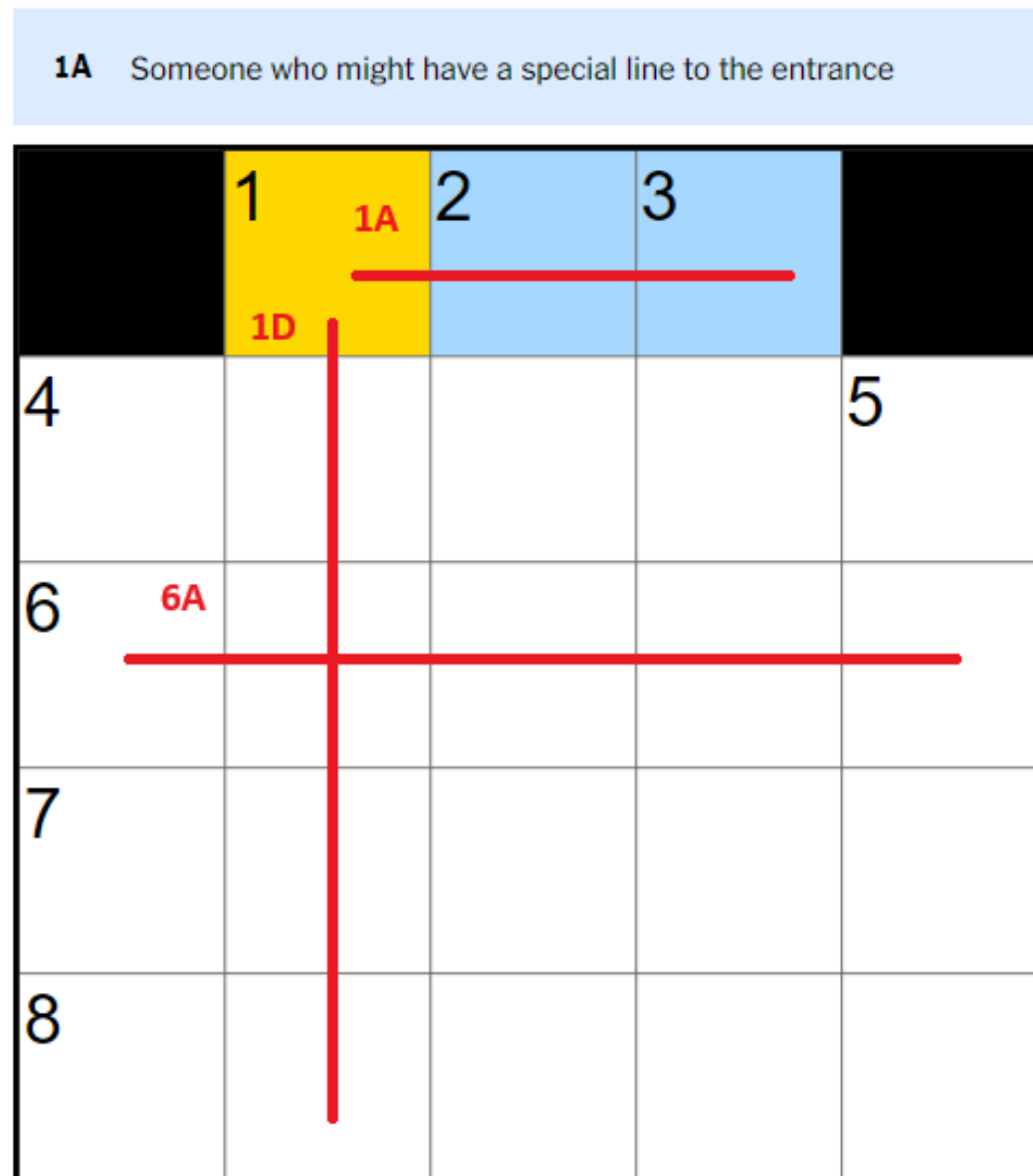
- 1 Crop sowing season (4)
- 4 Quintessential New Year gift (5)
- 8 Popular card game at house parties (3)
- 10 Bill waiting to be settled (3)
- 12 Say 'cheers' like a Viking (4)
- 14 What's common to a coat and a moat? (3)
- 16 A dish that is a mathematician's staple (3)
- 18 A common Goodreads challenge in French (2)
- 20 I sound like Sherlock's sister but I am a tipsy organic compound (4)
- 22 Kannada New Year (5)

Down

- 1 I am made and broken within weeks, sometimes days (10)
- 2 To stop illegal trading (4)
- 3 You could use me to write, paint, or modify your body (3)
- 5 Carry it to show who you are (2)
- 6 Gold in chemistry (2)
- 7 Some have a cause, some don't (6)
- 9 Bengaluru techies need it the most (3)
- 11 These can jam signals (10)
- 13 An excuse to avoid an outing (3)
- 15 By the Gregorian calendar, I will arrive after 212 days in 2023 (3)
- 17 Expertise of an otolaryngologist (3)
- 19 I sound like a Hawaiian but I hail from Nigeria. I am a close relative of the Indian percussion

Modelo de datos

Las celdas entonces tienen que estar etiquetadas



ACROSS

- 1 Someone who might have a special line to the entrance
- 1A** Someone who might have a special line to the entrance
- 4 Dial-up device
- 6 The "U" in UV
- 7 Feature of a well-used apron, maybe
- 8 Bird on the dollar bill

DOWN

- 1 Physicist for whom an electrical measurement is named
- 1D** electrical measurement is named
- 2 Dog collar attachment, for short
- 3 Danger
- 4 Artist's creative inspiration
- 5 Quite a head of hair

```

1 - {}
2 - "crossword": [
3 -   [
4 -     {
5 -       "dark": true,
6 -       "solution": "."
7 -     },
8 -     {
9 -       "dark": false,
10 -      "solution": "V",
11 -      "label": "1",
12 -      "across": "1A",
13 -      "down": "1D"
14 -     },
15 -     {
16 -       "dark": false,
17 -       "solution": "I",
18 -       "label": "2",
19 -       "across": "1A",
20 -       "down": "2D"
21 -     },
22 -     {
23 -       "dark": false,
24 -       "solution": "P",
25 -       "label": "3",
26 -       "across": "1A",
27 -       "down": "3D"
28 -     },
29 -     {
30 -       "dark": true,
31 -       "solution": "."
32 -     }
33 -   ]
34 - ]
35 - {}
  
```

Modelo de datos

Las modeladas pistas se ven así:

```

1 - {
2 -   "clues": {
3 -     "across": [
4 -       {
5 -         "label": "1A",
6 -         "hint": "Someone who might have a special line to the entrance"
7 -       }
8 -     ],
9 -     "down": [
10 -      {
11 -        "label": "1D",
12 -        "hint": "Physicist for whom an electrical measurement is named"
13 -      }
14 -    ]
15 -   }
16 - }

```



1A Someone who might have a special line to the entrance

	1	1A	2	3	
4	1D				5
6	6A				
7					
8					

ACROSS

1 Someone who might have a special line to the entrance
1A Someone who might have a special line to the entrance
4 Dial-up device
6 The "U" in UV
7 Feature of a well-used apron, maybe
8 Bird on the dollar bill

DOWN

1 Physicist for whom an electrical measurement is named
1D electrical measurement is named
2 Dog collar attachment, for short
3 Danger
4 Artist's creative inspiration
5 Quite a head of hair

Modelo de datos



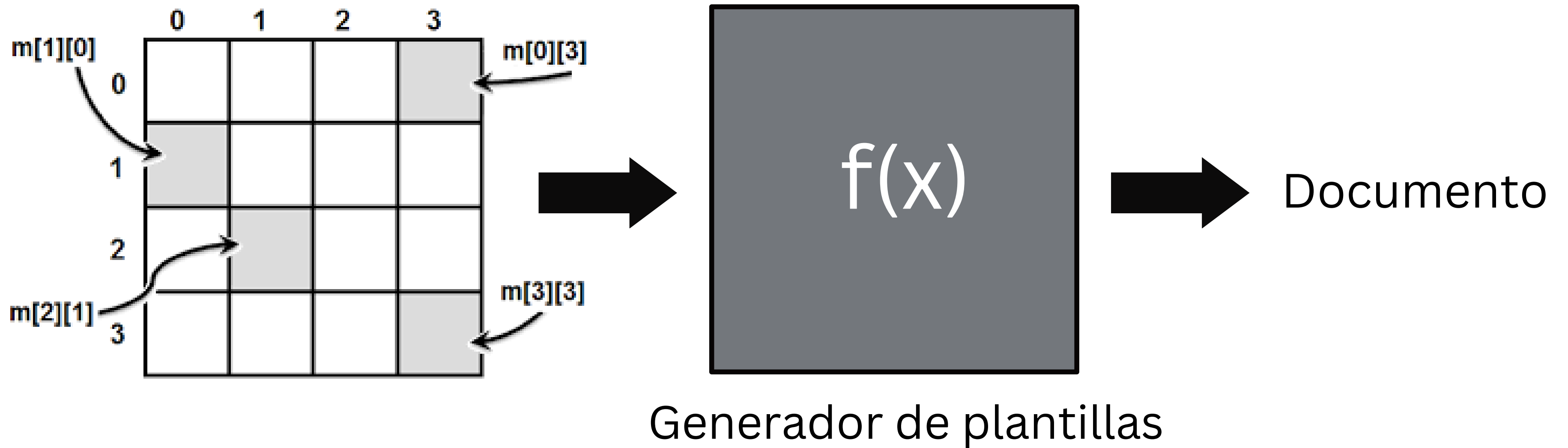
¿No es muy largo
el documento?
Como tenga que
hacerlo a mano...



Si, tienes razón.
Por eso he creado
un programa
que crea estas
plantillas

Modelo de datos

13



Tecnologías

¿Cómo has hecho la aplicación

La página web está hecha en React y el servidor lo he hecho con Express



Tecnologías

- En la aplicación puedes resolver crucigramas
- Y buscar palabras en un diccionario

Aquí se muestran las pistas. ¡Haz click en alguna!

1

2

3

4

5

6

7

8

9

10

Comprobar crucigrama

Buscar palabra en el diccionario

a?a

Página: 1/2

Anterior

Siguiente

Buscador inteligente

Si te falta alguna solución y tienes letras descubiertas búscala en el **diccionario de español** integrado.

Esta herramienta es similar en funcionalidad al buscador de [OneLook](#).

Formato de búsqueda

Horizontal

1A

Me desarrollé con facilidad, como agua en el río.

5A

Bebida tradicional de la India hecha a base de yogur.

7A

Conducirlo de un lugar a otro.

8A

Terminad concluid.

9A

Fibra que se obtiene de la planta de cannabis sativa.

10A

Mangos, asideros.

Vertical

1D

Las vacas, cuando hay crisis económica.

2D

Pasticho.

3D

Empleabas, utilizabas.

4D

Religión cuyo dogma se basa en el Corán.

6D

Yodo.

7D

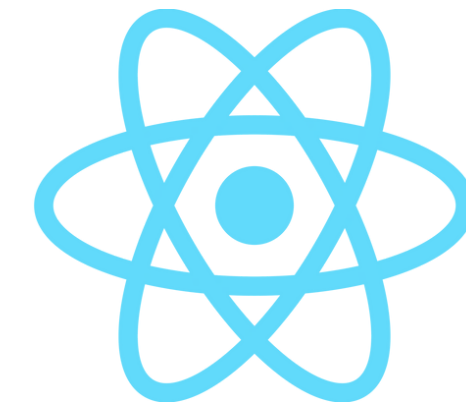
Fardo o lío.

13 palabras coinciden con a?a

Posición	Resultados de a?a
1	aba
2	aca
3	ada
4	aja
5	ala
6	ama

Conclusiones

Se ha desarrollado una página web



Se ha implementado una API

express

Se ha pasado del modelo conceptual a un documento



Conclusiones

Se ha explorado las agregaciones en MongoDB



Se ha automatizado automatizado tareas muy manuales

Se busca en un volumen grande de datos con Elasticsearch



Gracias por su atención

Pedro González Marcos

Grupo 34

Correo: **pedgonmar2@alum.us.es**

Repositorio: <https://github.com/pgmarc/crossword>